

Akshat Kalra

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EDUCATION

The University of British Columbia

BSc Computer Science and Statistics | Dean's Honor List

Vancouver, BC

Expected Graduation Nov 2027

- **Coursework:** Statistical Modeling (95%), Models of Computation (93%), Probability Theory (91%)
- **Awards:** 2x Faculty of Science International Student Scholarship (2025, 2024), OIS Scholarship

EXPERIENCE

Data Scientist Co-op, Atreides

Python, PySpark, SQL, Kubernetes, AWS

May 2026 – Present

Victoria, BC (Remote)

- **Source portfolio optimization.** Cut projected data spend **29%**, lifted coverage **85% to 99.4%**
- Modelled source selection as a **mixed-integer linear program** solved in **HiGHS** to a 0.1% gap
- Set binary source variables, coverage constraints, an estimated cost model, and dual objectives
- Scored candidates with a config-driven **PySpark** job over a **2.5 trillion-row** event table
- Productionized it as parameterized **Spark-on-Kubernetes** workflows writing to **Apache Iceberg**
- **Self-serve analytics.** Built an **MCP server** and **Grafana** layer over pre-aggregated metrics
- Let sales and non-technical staff query the event store in plain English, clearing a request queue
- Pre-computed bitmap-backed aggregates with scheduled **Spark** jobs, replacing full table scans
- **Anomaly detection.** Leading a reframe to **event-level classification** on **3.1M** labelled events

Software Developer Co-op, Faculty of Forestry, UBC

TypeScript (Next.js, Nest.js), Python, PostgreSQL (pgvector), AWS

May 2025 – April 2026

Vancouver, BC

- Built an **LLM evaluation pipeline** benchmarking providers against ground-truth instructor grades
- Measured agreement by **ICC**, **RMS error**, and **bootstrap CIs** to choose the production model
- Shipped a **pgvector** embeddings pipeline powering **semantic search** in HelpMe, used by **1,500+ students**
- Owned an AI grading assistant end to end on **AWS serverless** (Lambda, DynamoDB, API Gateway)
- Provisioned it as code via SAM/CloudFormation, deployed through **GitHub Actions CI/CD**

Undergraduate Teaching Assistant, UBC

Departments of Computer Science & Philosophy

Sep 2023 – Present

Vancouver, BC

- **CPSC 210** (current): software construction in TypeScript
- Taught **5 terms** of **CPSC 121**: discrete math, proofs, and intro to theory of computing
- **PHIL 220** (2 terms): symbolic logic and proof techniques

PROJECTS

Rustlock | Rust, Linux CLI, AES-256-GCM, Argon2id

Feb 2026

- Built a local-first **CLI password manager in Rust** with AES-256-GCM authenticated encryption
- Derived keys with Argon2id at a **64MB** memory cost to raise the cost of GPU brute-force attacks
- Generated passwords from OS entropy; prepended salt and nonce so vaults need no external metadata

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Rust, SQL (PostgreSQL, Trino/Athena), R

Data at Scale: Apache Spark (PySpark), Apache Iceberg, Trino, Athena, batch pipelines, ETL

Systems & Cloud: Linux, Docker, Kubernetes, Git, CI/CD (GitHub Actions), IaC, AWS (Lambda, S3, DynamoDB)

Statistics & ML: PyTorch, scikit-learn, NumPy, Pandas, classification, mixed-integer optimization (HiGHS)

AI Tooling: Claude Code, MCP servers, agentic workflows, LLM evaluation, vector search (pgvector)

Web: React.js, Next.js, Nest.js, Node.js, FastAPI, Grafana